FOUR DESCRIPTIONS OF A FINITE MODULE OVER A DOMAIN RELATIVE TO ITS RANK

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Rank over a domain, and the identification of M tensor K as a finite dimensional vector space for finite M. definition environment at source lines 5015-5019 of the same chapter. Supplies the r of the statement and the finite dimensional space in which the basis e_1 to e_r is chosen.
- A finitely generated torsion module over a domain is annihilated by a single nonzero element. used twice at proof lines 5120-5121 and 5126-5128, for the cokernel of the comparison map and for the finite torsion module N. Produces g and g' whose product f makes the kernel and cokernel of the comparison map vanish after localisation.
- A surjection onto a projective module splits. used at proof lines 5123-5124 to split M_g as N plus R_g^r. Isolates the finite torsion piece N whose annihilation gives the local freeness of M_f.

Source reference. Original lemma 0H9Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21544–21544. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *Let R be a domain. Let M be a finite R-module. Then*

(1) *M has finite rank $r \geq 0$,*
(2) *there is a map $M \to R^{\oplus r}$ whose kernel and cokernel are torsion modules,*

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(3) *there is an $f \in R$, $f \neq 0$ such that M_f is free of rank r, and*
(4) *there is an injective map $R^{\oplus r} \to M$ whose cokernel is a torsion module.*

Proof. Since M is finite, $M \otimes_R K$ is a finite dimensional vector space over the fraction field K of R. Thus the rank r is finite.

Choose generators $x_1, \ldots, x_n$ of M. Choose a basis $e_1, \ldots, e_r$ for $M \otimes_R K$. We may write $x_i \otimes 1 = \sum (a_{ij}/b_{ij}) e_j$ for some $a_{ij}, b_{ij} \in R$ with $b_{ij} \neq 0$. Setting $b = \prod b_{ij}$ and replacing a_{ij} by $a_{ij} b / b_{ij}$ we may assume that $x_i \otimes 1 = \sum (a_{ij}/b) e_j$. Note that if we have a relation $\sum r_i x_i = 0$ in M then $\sum r_i a_{ij} = 0$ in R for all $j = 1, \ldots, r$.

Proof of (2). The last comment above tells us the map $M \to \bigoplus Re_j$ sending x_i to $\sum a_{ij} e_j$. is well defined. The fact that $e_1, \ldots, e_r$ forms a basis of $M \otimes_R K$ tells us that this map becomes an isomorphism upon tensoring by K. Hence this is a map as in (2).

Proof of (3). The cokernel of the map $M \to R^{\oplus r}$ constructed in (2) is finite and torsion. Hence we can find a $g \in R$, $g \neq 0$ which annihilates this cokernel. In other words, such that the map $M_g \to R_g^{\oplus r}$ is surjective. Then we can choose a splitting $M_g = N \oplus R_g^{\oplus r}$. Then N is a finite R_g-module with $N \otimes_{R_g} K = 0$. In other words, N is torsion and finite over R_g. Thus we can find a $g' \in R$ which annihilates N. Setting $f = gg'$ we see that (3) holds.

To prove (4) choose a basis $y_1, \ldots, y_r$ of M_f where f is as in (3). Then write $y_j = m_j / f^{t_j}$ for some $m_j \in M$ and $t_j \geq 0$. The map $R^{\oplus r} \to M$ sending the jth basis vector to m_j will be as in (4). Details omitted. $\square$